

# Unifying Special Relativity and Quantum Mechanics

## Lecture 22 — Non-Abelian Gauge Theories

(Compact Lie groups, the covariant derivative, the adjoint representation, the field strength  $F_{\mu\nu}^A$ , and Yang–Mills theory)

Typed version of the handwritten notes, with added explanations in blue boxes

### How to read this document

**Conventions.** Exactly as in Lectures 1–21: normal text and the yellow / pink / grey boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, pencil). Everything inside a blue box like this one is material I have added; it is *not* in the original notes.

**Pagination.** This scan opens a *new* chapter, “Non-abelian gauge theories”, and the page numbering **restarts at 1** again. Since the previous chapter carried ten stars, these sheets are marked with eleven: [p. 1\*\*\*\*\*]–[p. 24\*\*\*\*\*]. This scan is in order and complete — no pages are swapped, duplicated or missing.

**Where the scan stops.** Sheet [p. 24\*\*\*\*\*] opens the case of semi-simple groups and breaks off just after stating that  $G_\mu^A$  and  $F_{\mu\nu}^A$  again transform in the adjoint. A closing blue box completes that thought — in particular the one point the construction implies but the sheet does not reach: a semi-simple group has *one independent coupling per simple factor*.

**Notation.** The gauge fields are written  $G_\mu^A$  throughout (not  $A_\mu^a$ ), with capital latin  $A, B, C = 1, \dots, \dim G$  labelling the generators and lower-case latin  $a, b$  labelling components of the matter multiplet. Do not confuse the two alphabets:  $\Phi^a$  carries a *representation* index,  $G_\mu^A$  an *algebra* index.

**One thing to watch.** The coupling  $g$  appears only from [p. 17\*\*\*\*\*] onward, and its placement changes at [p. 18\*\*\*\*\*] when the fields are rescaled  $G_\mu^A \rightarrow g G_\mu^A$ . Before that rescaling the Lagrangian carries  $-\frac{1}{4g^2} F_{\mu\nu}^A F^{A,\mu\nu}$  with  $g$  absent from  $D_\mu$  and from  $F$ ; afterwards it carries  $-\frac{1}{4} F_{\mu\nu}^A F^{A,\mu\nu}$  with  $g$  present in both. Both conventions appear in the notes, a few lines apart, and they describe the same theory.

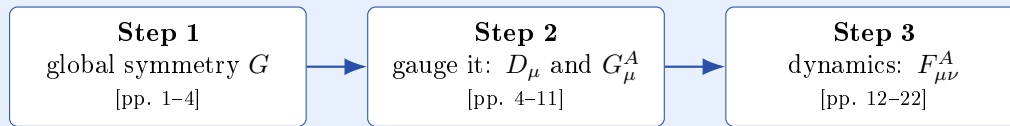
### Where we left off, and where this is going

The gravity chapter is closed. Lectures 16–19 built a massless spin-2 field and found that consistency forced general relativity; Lectures 20–21 coupled matter to it and found that the resulting quantum theory is an effective theory with cutoff  $M_P$ . Along the way, one technique kept reappearing: take a *global* symmetry, make its parameter depend on  $x$ , and discover that a new field — and its interactions — are forced on you. That was how QED was built, and it was how the graviton’s gauge symmetry was deformed into general coordinate invariance in Lecture 19.

This lecture applies the same machine one more time, in its most consequential form. The starting symmetry is now a *non-abelian* compact Lie group, and the output is Yang–Mills theory: the framework underlying both the strong interaction ( $SU(3)$ ) and the electroweak theory ( $SU(2) \times U(1)$ ).

The single new feature, compared with QED, is that the gauge fields themselves carry charge. In electromagnetism the photon is electrically neutral and the theory is linear; here the gauge bosons transform in the adjoint representation, and the theory acquires cubic and quartic

self-interactions. Structurally this is the same phenomenon that made gravity non-linear in Lecture 19 — with the crucial difference, established at the end, that here the coupling is *dimensionless*, so the theory is renormalizable and the non-linearities stop at quartic order. The construction proceeds in three steps, labelled in the notes:



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## STEP 1 — A global symmetry under a compact Lie group (pp. 1\*\*\*\*\*-4\*\*\*\*\*)

### 1 Global symmetry and compact Lie groups [p. 1\*\*\*\*\*]

[p. 1\*\*\*\*\*]

#### ■ Non-abelian gauge theories

Motivated by the success of the minimal coupling prescription, we try to construct a non-abelian generalization of electromagnetism

Step 1 We start considering a free Lagrangian that is invariant under a global symmetry group  $G$  that is a compact Lie group

$\mathcal{L} = \mathcal{L}(\Phi^a, \partial_\mu \Phi^a)$  quadratic in the field, with at most two derivatives, invariant under the transformation

$$\Phi^a(x) \longrightarrow \Phi'^a(x) = \left[ \exp(-i\theta_A T^A) \right]^a_b \Phi^b(x)$$

$\nearrow \theta_A$  are the global parameters that identify a transformation of  $G$

$\nwarrow T^A, A = 1, \dots, \dim G$  are the generators

The generators close the Lie algebra

$$[T^A, T^B] = if_{ABC} T^C$$

$\nwarrow$  “structure constants”  
fully antisymmetric if the group is compact

#### Unpacking the starting point

Three requirements are being imposed at once, and each earns its place:

“**Quadratic in the field, at most two derivatives**” — this is the free Lagrangian. Restricting to two derivatives keeps the theory renormalizable; quadratic means no interactions yet.

“**Global**” — the parameters  $\theta_A$  are constants, the same at every point of spacetime. This is the symmetry that Step 2 will localize.

“**Compact**” — this is doing more work than it looks. Compactness guarantees (i) that finite-dimensional representations can be taken unitary, hence the generators Hermitian (next page); (ii) that the structure constants can be chosen *fully antisymmetric*, which is what makes  $\text{tr}(T^A T^B) \propto \delta^{AB}$  and hence lets  $F_{\mu\nu}^A F^{A,\mu\nu}$  be a sensible, positive-definite kinetic term later on. A non-compact group would give kinetic terms with the wrong sign for some generators — ghosts.

**Reading the exponential.**  $\exp(-i\theta_A T^A)$  is a matrix acting on the index  $a$  of the multiplet  $\Phi^a$ ; the sum over  $A$  runs over the  $\dim G$  generators. “Non-abelian” means the  $T^A$  do not commute, and the Lie algebra  $[T^A, T^B] = if_{ABC} T^C$  quantifies exactly how much they fail to. In electromagnetism  $G = U(1)$  has one generator,  $f_{ABC} = 0$ , and everything below collapses to the familiar abelian case. *Every* new feature of this lecture traces back to  $f_{ABC} \neq 0$ .

## 2 Hermitian generators; the simple compact groups [p. 2\*\*\*\*\*]

[p. 2\*\*\*\*\*]

The generators are Hermitian,

$$(T^A)^\dagger = T^A$$

since  $G$  is compact, finite dim. representations are equivalent to unitary representations

We also write in compact form

$$\Phi(x) \rightarrow U(\vec{\theta}) \Phi(x), \quad U(\vec{\theta}) = \exp(-i\theta_A T^A)$$

We start by focusing on compact simple Lie group

► non-abelian, with no non-trivial normal subgroup

These groups are a sort of “elementary objects” and can be classified

| $G$                            | $\dim G$               |                      |
|--------------------------------|------------------------|----------------------|
| $SU(N)$                        | $N^2 - 1$              |                      |
| $SO(N)$                        | $N(N - 1)/2$           |                      |
| $Sp(N)$                        | $N(2N + 1)/2$          |                      |
| $E_6, E_7, E_8,$<br>$F_4, G_2$ | 78, 133, 248<br>52, 14 | “exceptional groups” |

### Why Hermitian, and what the table is

**Hermiticity.** If the representation is unitary,  $U^\dagger U = 1$ . Writing  $U = \exp(-i\theta_A T^A)$  and expanding to first order,  $U^\dagger U = (1 + i\theta_A (T^A)^\dagger)(1 - i\theta_B T^B) = 1 + i\theta_A [(T^A)^\dagger - T^A] + \mathcal{O}(\theta^2)$ , so unitarity at first order is  $(T^A)^\dagger = T^A$ . Compactness is what guarantees a unitary representation exists in the first place (one can average over the group with an invariant measure of finite volume — Weyl’s unitarian trick).

**The classification.** This is the Cartan classification of simple compact Lie algebras, and its content is that the table is *complete*: four infinite families plus exactly five exceptional cases, and nothing else. It is one of the great results of nineteenth-century mathematics and it is why particle physics has such a short menu of candidate gauge groups.

**Sanity checks on the dimensions.**  $SU(N)$  matrices are  $N \times N$ , unitary ( $N^2$  real constraints on  $2N^2$  real parameters, leaving  $N^2$ ) and of unit determinant (one more), so  $N^2 - 1$ . For  $SU(2)$  this gives 3 (the Pauli matrices), for  $SU(3)$  it gives 8 (the eight gluons).  $SO(N)$  is the number of independent rotation planes,  $N(N - 1)/2$ : for  $N = 3$ , three rotations.

**Where these appear in nature.**  $SU(3)$  is the colour group of QCD —  $\dim = 8$ , hence eight gluons.  $SU(2) \times U(1)$  is the electroweak group. The exceptional groups  $E_6$  and  $E_8$  appear in grand-unification and string model-building.

## 3 Example: $N$ Dirac fields in the fundamental [pp. 3–4\*\*\*\*\*]

[p. 3\*\*\*\*\*]

The explicit form of the generators depends on the representation of  $G$  under which the fields  $\Phi^a$  transform.

Example

Consider a set of  $N$  Dirac field  $\psi^a(x)$ ,  $a = 1, \dots, N$

We collect them into a  $N$ -dimensional vector

$$\Psi(x) = \begin{pmatrix} \psi^1(x) \\ \vdots \\ \psi^N(x) \end{pmatrix}$$

← N.B. each  $\psi^a(x)$  has, in turn, 4 components since it is a Dirac spinor

that transform under the group  $SU(N)$  according to the fundamental representation

$$\Psi(x) \longrightarrow \Psi'(x) = \exp\left(-i \sum_{A=1}^{N^2-1} \theta_A T^A\right) \Psi(x) = U(\vec{\theta}) \Psi(x)$$

We also introduce

$$\bar{\Psi}(x) = (\bar{\psi}_1(x), \dots, \bar{\psi}_N(x))$$

$$\begin{aligned} \Psi^\dagger &= (\psi^{1\dagger}, \dots, \psi^{N\dagger}) \\ \bar{\Psi} &= \Psi^\dagger \gamma^0 \mathbf{1}_{N \times N} = \bar{\Psi}(x) \end{aligned}$$

that transforms according to

$$\bar{\Psi}(x) \longrightarrow \bar{\Psi}(x) U(\vec{\theta})^{-1}$$

The Lagrangian

$$\mathcal{L} = \bar{\Psi} i \gamma^\mu (\partial_\mu \Psi) - m \bar{\Psi} \Psi$$

is invariant under the global  $SU(N)$  group

✓ all fields have the same mass “ $m$ ”

Since the symmetry is global

$$\partial_\mu \Psi(x) \longrightarrow \partial_\mu (U(\vec{\theta}) \Psi(x)) = U(\vec{\theta}) (\partial_\mu \Psi(x))$$

The term

$$\bar{\Psi} i \gamma^\mu (\partial_\mu \Psi) \longrightarrow \bar{\Psi} U^{-1} i \gamma^\mu U (\partial_\mu \Psi), \quad \text{therefore, is trivially invariant.}$$

### Two index structures at once, and why the mass is universal

**Two kinds of index.**  $\Psi$  carries a Dirac index (4 components, acted on by  $\gamma^\mu$ ) *and* a group index (N components, acted on by  $T^A$  and  $U$ ). The two never mix:  $\gamma^\mu$  is proportional to the identity in group space,  $U$  is proportional to the identity in spinor space. That is exactly why  $U^{-1} \gamma^\mu U = \gamma^\mu$  in the line above — they commute because they act on different indices. The explicit  $\mathbf{1}_{N \times N}$  in the definition of  $\bar{\Psi}$  is the note making this point.

**Why  $\bar{\Psi} \rightarrow \bar{\Psi} U^{-1}$ .** From  $\Psi \rightarrow U \Psi$  we get  $\Psi^\dagger \rightarrow \Psi^\dagger U^\dagger$ , and  $U$  unitary gives  $U^\dagger = U^{-1}$ . So  $\bar{\Psi} \Psi \rightarrow \bar{\Psi} U^{-1} U \Psi = \bar{\Psi} \Psi$ : the mass term is invariant.

**Why all  $N$  fields must share one mass  $m$ .** The mass term  $m \bar{\Psi} \Psi = m \sum_a \bar{\psi}_a \psi^a$  is only invariant because it is proportional to the identity in group space. Different masses would mean  $\bar{\Psi} M \Psi$  with  $M = \text{diag}(m_1, \dots, m_N)$ , and invariance would require  $U^{-1} M U = M$  for

[p. 4\*\*\*\*\*]

every  $U \in SU(N)$  — which forces  $M \propto \mathbf{1}$ . So an exact non-abelian symmetry *predicts* a degenerate multiplet. This is why isospin symmetry (approximate  $SU(2)$ ) goes with the near-degeneracy of the proton and neutron, and why breaking such symmetries is what splits masses.

## STEP 2 — Gauging the symmetry (pp. 4\*\*\*\*\*–11\*\*\*\*\*)

### 4 The problematic term [pp. 4–5\*\*\*\*\*]

Step 2 We “gauge” the symmetry, and promote  $\vec{\theta}$  to be a space-time dependent set of parameters

$$\Phi^a(x) \longrightarrow \Phi'^a(x) = \underbrace{\exp\left(-i\theta_A(x)T^A\right)^a_b}_{\text{problematic term}} \Phi^b(x)$$

$$\hookrightarrow \equiv U(\vec{\theta}(x)) \equiv U(x)$$

The Lagrangian  $\mathcal{L}(\Phi^a, \partial_\mu \Phi^a)$  is no longer invariant, and the reason is the derivative term

Consider the previous example for illustration.

[p. 5\*\*\*\*\*]

$$\begin{aligned} \partial_\mu \Psi(x) &\longrightarrow \partial_\mu \left( \exp\left(-i\theta_A(x)T^A\right) \Psi(x) \right) = \partial_\mu (U(x) \Psi(x)) \\ &= \underbrace{(\partial_\mu U(x)) \Psi(x)}_{\text{problematic term}} + U(x) (\partial_\mu \Psi(x)) \end{aligned}$$

We try to solve the issue by following the same logic of the minimal coupling prescription: we introduce a “covariant derivative”

$$D_\mu \Phi = \partial_\mu \Phi + i G_\mu^A T^A \Phi$$

► we introduce  $\dim G$  “gauge fields”

and we impose that this covariant derivative is such that it transforms under the local transformation as the ordinary derivative does when the transformation is local

$$D_\mu \Phi \xrightarrow[G]{\text{local}} U(x) (D_\mu \Phi)$$



Let’s work out the implications of this condition.

Let’s compute both sides of the transformation

$$(D_\mu \Phi)' \stackrel{!}{=} U(x) (D_\mu \Phi)$$

**The idea in one sentence**

The ordinary derivative fails because it differentiates  $U(x)$  as well as  $\Phi(x)$ , producing the stray term  $(\partial_\mu U)\Phi$ . The fix is to add a new field whose transformation law is *designed* to cancel exactly that stray term.

This is not a trick invented here — it is the same move as in QED ( $D_\mu = \partial_\mu + iqA_\mu$ ), and structurally the same as the covariant derivative of general relativity, where the Christoffel symbols cancel the inhomogeneous piece in the transformation of  $\partial_\mu V^\nu$ . In all three cases:

|            | stray term                             | compensating field         |
|------------|----------------------------------------|----------------------------|
| QED        | $\partial_\mu \alpha$                  | $A_\mu$ (1 field)          |
| Yang–Mills | $(\partial_\mu U)U^{-1}$               | $G_\mu^A$ (dim $G$ fields) |
| gravity    | $\partial_\mu(\partial x'/\partial x)$ | $\Gamma_{\mu\nu}^\lambda$  |

Note the counting: *one gauge field per generator*.  $SU(3)$  has eight generators, hence eight gluons. Nothing chooses this number by hand; it is  $\dim G$ .

**5 Deriving the gauge-field transformation law [p. 6\*\*\*\*\*]**

[p. 6\*\*\*\*\*]

On the left side:

$$\begin{aligned}
 (D_\mu \Phi)' &= (\partial_\mu \Phi + iG_\mu^A T^A \Phi)' = \partial_\mu \Phi' + iG_\mu'^A T^A \Phi' = \\
 &= \partial_\mu (U\Phi) + iG_\mu'^A T^A U\Phi = \\
 &= (\partial_\mu U)\Phi + U(\partial_\mu \Phi) + iG_\mu'^A T^A U\Phi
 \end{aligned}$$

On the right side:

$$U(x)(D_\mu \Phi) = U(\partial_\mu \Phi + iG_\mu^A T^A \Phi) = U(\partial_\mu \Phi) + iU G_\mu^A T^A \Phi$$

■ We now compare

$$\begin{aligned}
 (\partial_\mu U)\Phi + \cancel{U(\partial_\mu \Phi)} + iG_\mu'^A T^A U\Phi &= \cancel{U(\partial_\mu \Phi)} + iU G_\mu^A T^A \Phi \\
 iG_\mu'^A T^A U\Phi &= iU G_\mu^A T^A \Phi - (\partial_\mu U)\Phi
 \end{aligned}$$

at the level of operators

$$G_\mu'^A T^A U = U G_\mu^A T^A + i(\partial_\mu U)$$

multiplying by  $U^{-1}$  on the right

$$G_\mu'^A T^A = i(\partial_\mu U)U^{-1} + U G_\mu^A T^A U^{-1}$$

This computation shows that if

$$G_\mu^A T^A \longrightarrow G_\mu'^A T^A = i(\partial_\mu U)U^{-1} + U G_\mu^A T^A U^{-1}$$

[p. 7\*\*\*\*\*]

then the covariant derivative transforms as in ★

↔ The non-abelian analogue to the abelian combined gauge transformations is

$$\begin{cases} \Phi(x) \longrightarrow U(x)\Phi(x) \\ G_\mu^A(x)T^A \longrightarrow i[\partial_\mu U(x)]U(x)^{-1} + U(x)G_\mu^A(x)T^AU(x)^{-1} \end{cases}$$

### Reading the two pieces of the transformation

The gauge-field transformation splits into two structurally different terms, and almost everything interesting in this lecture follows from the split:

$$G_\mu \longrightarrow \underbrace{i(\partial_\mu U)U^{-1}}_{\text{inhomogeneous}} + \underbrace{U G_\mu U^{-1}}_{\text{homogeneous}}, \quad G_\mu \equiv G_\mu^A T^A.$$

- The **inhomogeneous** term is the shift that cancels  $(\partial_\mu U)\Phi$ . It survives only when  $U$  depends on  $x$ , so it encodes the *local* nature of the symmetry. In QED it is the familiar  $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$ .
- The **homogeneous** term is a conjugation. It is present *even for constant*  $U$ , and it says the gauge fields carry a group index that gets rotated. **This term has no analogue in QED**: for  $U(1)$ ,  $U$  is a number, conjugation does nothing, and  $A_\mu$  is inert under a global phase rotation.

Pages 8–12 are devoted entirely to unpacking the second term, because it is what makes the theory non-abelian.

A note on the compact matrix notation: writing  $G_\mu \equiv G_\mu^A T^A$  turns the whole set of  $\dim G$  fields into a single matrix-valued object, and lets  $U(\dots)U^{-1}$  be written without indices. It is worth getting comfortable with, since every subsequent manipulation uses it.

## 6 The gauged Dirac Lagrangian [p. 7\*\*\*\*\*]

Consequently, if we promote the ordinary derivative to be covariant, we get a theory that is invariant under the “gauged” symmetry

### Example

Consider the Lagrangian we constructed before

$$\mathcal{L} = \bar{\Psi} i\gamma^\mu \partial_\mu \Psi - m\bar{\Psi}\Psi$$

↓ if we promote the derivative

$$\begin{aligned} \mathcal{L} &= \bar{\Psi} i\gamma^\mu (D_\mu \Psi) - m\bar{\Psi}\Psi \\ D_\mu \Psi &= \partial_\mu \Psi + i G_\mu^A T^A \Psi \end{aligned}$$

invariant under local  $SU(N)$  symmetry

The kinetic term is now invariant since

$$\bar{\Psi} i\gamma^\mu D_\mu \Psi \longrightarrow \bar{\Psi} U^{-1} i\gamma^\mu U (D_\mu \Psi) = \bar{\Psi} i\gamma^\mu (D_\mu \Psi)$$

Before moving to Step 3, let us discuss a bit more the transformation rule of the gauge fields.

## 7 The global limit [pp. 8–9\*\*\*\*\*]

[p. 8\*\*\*\*\*]

$$G_\mu^A T^A \longrightarrow i(\partial_\mu U)U^{-1} + U G_\mu^A T^A U^{-1}$$

(compared to the abelian  $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$ )

Let's discuss the above transformation by considering, as a specific example, the case of the Dirac  $SU(N)$  model.

We write explicitly the Lagrangian as

$$\begin{aligned}\mathcal{L} &= \bar{\Psi} i \gamma^\mu (\partial_\mu \Psi + i G_\mu^A T^A \Psi) - m \bar{\Psi} \Psi = \\ &= \bar{\Psi} i \gamma^\mu (\partial_\mu \Psi) - m \bar{\Psi} \Psi - \bar{\Psi} \gamma^\mu G_\mu^A T^A \Psi\end{aligned}$$

$$\mathcal{L} = \bar{\Psi} i \gamma^\mu (\partial_\mu \Psi) - m \bar{\Psi} \Psi - \bar{\Psi} \gamma^\mu G_\mu^A T^A \Psi$$

By construction, this Lagrangian is invariant under a local gauge transformation.

Consider now the limit in which the transformation is global,  $\theta_A(x) \equiv \theta_A = \text{const.}$

The Lagrangian must be invariant, since a global transformation is a special case of a local one.

If we “downgrade” the symmetry to be global, the transformations are

$$\Psi(x) \longrightarrow U \Psi(x); \quad G_\mu^A T^A \longrightarrow U G_\mu^A T^A U^{-1}$$

[p. 9\*\*\*\*\*]

The Lagrangian is obviously invariant.

In particular, the last term transforms according to

$$\begin{aligned}\bar{\Psi} \gamma^\mu G_\mu^A T^A \Psi &\longrightarrow \bar{\Psi} U^{-1} \gamma^\mu (U G_\mu^A T^A U^{-1}) U \Psi = \\ &= \bar{\Psi} \gamma^\mu G_\mu^A T^A \Psi, \quad \text{invariant}\end{aligned}$$

Consequently, in the transformation

$$G_\mu^A T^A \longrightarrow \underbrace{i(\partial_\mu U)U^{-1}}_{(i)} + \underbrace{U G_\mu^A T^A U^{-1}}_{(ii)}$$

(i) This term takes into account the local nature of the transformation when the global symmetry is gauged ( $\partial_\mu U \neq 0$ )

(ii) This term takes into account that, under a gauge transformation, the gauge fields transform non trivially

**△ Crucial difference** compared to electromagnetism, in which the photon field does not transform when the symmetry  $U(1)$  is global:  $A_\mu \rightarrow A_\mu$

We now ask the following, related, question. According to which representation of  $G$  the gauge field transform?

### The physical content: gauge bosons are charged

$A_\mu \rightarrow A_\mu$  under a global  $U(1)$  means the photon carries no electric charge.  $G_\mu^A \rightarrow U G_\mu^A T^A U^{-1}$  means the gauge bosons *do* carry the charge of the group: gluons are coloured.

The consequences are the whole difference between QED and QCD:

|                         | QED                   | Yang–Mills                   |
|-------------------------|-----------------------|------------------------------|
| gauge boson charge      | neutral               | charged (adjoint)            |
| self-interaction        | none                  | cubic and quartic vertices   |
| field equations         | linear                | non-linear                   |
| long-distance behaviour | Coulomb, free charges | confinement                  |
| running coupling        | grows with energy     | shrinks (asymptotic freedom) |

The same logical pattern appeared in Lecture 19: the graviton carries energy, energy is the gravitational charge, so gravity self-interacts. Here the gauge boson carries colour, colour is the charge, so the gluon self-interacts. Both theories are non-linear for the same structural reason — and the question of *which* representation the gauge field falls into is answered next.

## 8 Infinitesimal form: the adjoint representation [pp. 10–12\*\*\*\*\*]

[p. 10\*\*\*\*\*]

In general, under a global symmetry transformation, we have that a field transform according to (for an infinitesimal transformation)

$$\Phi^a(x) \longrightarrow \Phi'^a(x) = \Phi^a(x) - i\alpha_A (T^A)^a_b \Phi^b(x) + \mathcal{O}(\alpha^2)$$

In the infinitesimal form, we can isolate the generators of  $G$  in the representation according to which  $\Phi^a$  transform.

Let us thus write

$$G_\mu^A T^A \longrightarrow G_\mu'^A T^A = i(\partial_\mu U)U^{-1} + U(G_\mu^A T^A)U^{-1}$$

In infinitesimal form

$$U = \exp(-i\theta_A T^A) = \mathbf{1} - i\theta_A T^A + \mathcal{O}(\theta^2)$$

$$\begin{aligned} G_\mu'^A T^A &= i\left(\partial_\mu(\mathbf{1} - i\theta_A T^A)\right)(\mathbf{1} + i\theta_A T^A) + (\mathbf{1} - i\theta_A T^A)(G_\mu^B T^B)(\mathbf{1} + i\theta_A T^A) \\ &= i(-i)(\partial_\mu \theta_A)T^A + G_\mu^A T^A - i\theta_A T^A G_\mu^B T^B + iG_\mu^B T^B \theta_A T^A + \mathcal{O}(\theta^2) \\ &= (\partial_\mu \theta_A)T^A + G_\mu^A T^A - i\theta_A G_\mu^B [T^A, T^B] = \\ &= G_\mu^A T^A + (\partial_\mu \theta_A)T^A - i\theta_A G_\mu^B i f_{ABC} T^C = \\ &= G_\mu^A T^A + (\partial_\mu \theta_A)T^A + \theta_B G_\mu^C f_{BCA} T^A = \end{aligned}$$

[p. 11\*\*\*\*\*]

$$= \left(G_\mu^A + (\partial_\mu \theta_A) + f_{ABC} \theta_B G_\mu^C\right) T^A$$

We write

$$G_\mu^A \longrightarrow G_\mu'^A = G_\mu^A + \partial_\mu \theta^A + f_{ABC} \theta_B G_\mu^C$$

We elaborate on this expression as follows

$$\begin{aligned} G_\mu'^A &= G_\mu^A + \partial_\mu \theta^A - i(i f_{ABC}) \theta_B G_\mu^C \\ &= G_\mu^A + \partial_\mu \theta^A - i \underbrace{(-i f_{BAC})}_{f_{ABC}} \theta_B G_\mu^C \end{aligned}$$

↪ we define

So that

infinitesimal gauge field transformation

$$G_\mu^A \longrightarrow G_\mu'^A = G_\mu^A + (\partial_\mu \theta^A) - i\theta_B (t^B)_{AC} G_\mu^C$$

If the symmetry is global,  $\partial_\mu \theta^A = 0$ , and

$$G_\mu^A \xrightarrow[G]{\text{global}} G_\mu^A - i\theta_B (t^B)_{AC} G_\mu^C$$

We thus find that the gauge fields transform under (global)  $G$  according to the representation in which the generators take the form

[p. 12\*\*\*\*\*]

$$(t^B)_{AC} = -if_{BAC}$$

This is known as the “adjoint representation”

#### What the adjoint representation is, and why it had to be this one

**The definition.** An ordinary (say fundamental) representation assigns to each generator a matrix acting on the multiplet index  $a$ . The *adjoint* representation is the one in which the generators are given by the structure constants themselves,

$$(t^B)_{AC} = -if_{BAC},$$

so the matrices are  $\dim G \times \dim G$  and the “multiplet” being acted on is the algebra itself. Its dimension therefore equals the dimension of the group: for  $SU(3)$ , eight gluons in an eight-dimensional representation.

**Why it must be the adjoint.**  $G_\mu^A$  has exactly one index of the same type as  $\theta_A$  — an algebra index. There is only one natural way the algebra can act on itself, namely by the commutator, and that action is by definition the adjoint. The compact-form statement  $G_\mu \rightarrow U G_\mu U^{-1}$  says the same thing without indices: conjugation is the adjoint action.

**Check that these matrices really satisfy the algebra.** They do, and the proof is the Jacobi identity  $f_{ABE}f_{ECD} + f_{BCE}f_{EAD} + f_{CAE}f_{EBD} = 0$ , which is precisely the statement  $[t^A, t^B] = if_{ABC}t^C$ . So the structure constants are not merely bookkeeping — they are themselves a representation.

**The physical statement.** The gauge bosons transform in the adjoint; matter fields transform in whatever representation you chose for them (fundamental, in the example). In QCD: quarks in the fundamental (3 colours), gluons in the adjoint (8 colours). For  $U(1)$ ,  $f_{ABC} = 0$ , the adjoint is trivial, and the photon is neutral — which is exactly the earlier observation  $A_\mu \rightarrow A_\mu$ , now derived rather than noted.

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## STEP 3 — The dynamics of the gauge fields (pp. 12\*\*\*\*\*–22\*\*\*\*\*)

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### 9 Inspiration from QED: the commutator of covariant derivatives [p. 12\*\*\*\*\*]

Step 3 We introduce a term describing the dynamics of the gauge fields

We need a term analogue to  $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$  of QED.

Such would-be term must be:

- i) a function of  $G_\mu^A$  and derivatives;
- ii) gauge invariant.

↪ Let's take inspiration from QED

$$\begin{aligned} D_\mu D_\nu \psi &= (\partial_\mu + iqA_\mu)(\partial_\nu + iqA_\nu)\psi = \\ &= (\partial_\mu + iqA_\mu)(\partial_\nu \psi + iqA_\nu \psi) = \\ &= (\partial_\mu \partial_\nu \psi) + iq(\partial_\mu A_\nu)\psi + iqA_\nu(\partial_\mu \psi) + iqA_\mu(\partial_\nu \psi) - q^2 A_\mu A_\nu \psi \end{aligned}$$

If we take the difference  $D_\mu D_\nu \psi - D_\nu D_\mu \psi$ , all symmetric terms vanish, and we find

$$D_\mu D_\nu \psi - D_\nu D_\mu \psi = iq F_{\mu\nu} \psi$$

#### Why the commutator is the right thing to look at

Individually,  $D_\mu D_\nu \psi$  contains  $\partial_\mu \partial_\nu \psi$  and terms with derivatives acting on  $\psi$  — it depends on the field  $\psi$  in a complicated way. In the *commutator* every term that differentiates  $\psi$  is symmetric in  $\mu \leftrightarrow \nu$  and cancels, leaving something proportional to  $\psi$  with no derivatives on it at all. What multiplies  $\psi$  is therefore a pure function of the gauge field: the field strength. Two ways to say what this means:

- *Algebraically*:  $F_{\mu\nu}$  measures the failure of covariant derivatives to commute.
- *Geometrically*: it measures the failure of parallel transport around a closed infinitesimal loop to return you to where you started — it is a *curvature*. The identical statement in general relativity is  $[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho_{\sigma\mu\nu}V^\sigma$ , with the Riemann tensor in place of  $F_{\mu\nu}$ .

And crucially, this definition will make the transformation law of  $F$  easy to find — which is the point of the next page, and the reason for defining  $F$  this way rather than just guessing  $\partial_\mu G_\nu - \partial_\nu G_\mu$ .

## 10 Defining $F_{\mu\nu}^A$ [pp. 13–14\*\*\*\*\*]

[p. 13\*\*\*\*\*]

We mimic this equation, and look for a function  $F_{\mu\nu}^A(x)$  such that

$$D_\mu D_\nu \Phi - D_\nu D_\mu \Phi = i F_{\mu\nu}^A(x) T^A \Phi$$

■

■ First of all, we note that a function defined this way will have very “nice” transformation properties under local  $G$ .

↪ We work out the transformation properties of  $F_{\mu\nu}^A$

We know that

$$D_\mu \Phi \longrightarrow (D_\mu \Phi)' = U(D_\mu \Phi)$$

We also write  $(D_\mu \Phi)' = D'_\mu \Phi' = D'_\mu U \Phi$  so that, at operator level, we have

$$D'_\mu U \Phi = U D_\mu \Phi \quad \rightsquigarrow \quad \boxed{D_\mu \longrightarrow D'_\mu = U D_\mu U^{-1}}$$

As a consequence

$$D_\mu D_\nu \Phi \longrightarrow D'_\mu D'_\nu \Phi' = (U D_\mu U^{-1})(U D_\nu U^{-1}) U \Phi = U(D_\mu D_\nu \Phi)$$

We thus find that

$$D_\mu D_\nu \Phi - D_\nu D_\mu \Phi \xrightarrow[G]{\text{local}} U(D_\mu D_\nu \Phi - D_\nu D_\mu \Phi)$$

[p. 14\*\*\*\*\*]

From ■, we get:

$$\begin{aligned} i F_{\mu\nu}^A T^A \Phi &\xrightarrow[G]{\text{local}} i F_{\mu\nu}'^A T^A \Phi' = (D_\mu D_\nu \Phi - D_\nu D_\mu \Phi)' \\ &= U(D_\mu D_\nu \Phi - D_\nu D_\mu \Phi) \\ &= U i F_{\mu\nu}^A T^A \Phi \end{aligned}$$

We find

$$F_{\mu\nu}'^A T^A U \Phi = U F_{\mu\nu}^A T^A \Phi \xrightarrow[\text{level}]{\text{operator}} F_{\mu\nu}'^A T^A U = U F_{\mu\nu}^A T^A$$

that is  $F_{\mu\nu}'^A T^A = U(F_{\mu\nu}^A T^A)U^{-1}$

We find

$$F_{\mu\nu}^A T^A \xrightarrow[G]{\text{local}} F_{\mu\nu}'^A T^A = U(F_{\mu\nu}^A T^A)U^{-1}$$

↑ This is the same transformation rule of the global part of  $G_\mu^A T^A$

$F_{\mu\nu}^A$  transform according to the adjoint representation of  $G$

This is “nice” because it is simple to construct invariant objects.

Take  $\text{tr} \left[ F_{\mu\nu}^A T^A F^{B,\mu\nu} T^B \right]$

**The key structural point**

Compare the two transformation laws:

$$G_\mu \longrightarrow \underbrace{i(\partial_\mu U)U^{-1}}_{\text{inhomogeneous}} + UG_\mu U^{-1}, \quad F_{\mu\nu} \longrightarrow UF_{\mu\nu}U^{-1}.$$

$F$  transforms *homogeneously* — no inhomogeneous piece. That is what makes it usable for building a Lagrangian: an object that transforms by conjugation can be combined into invariants, whereas one with an additive shift cannot.

**But note the contrast with QED.** There,  $F_{\mu\nu}$  is fully *gauge invariant*:  $F_{\mu\nu} \rightarrow F_{\mu\nu}$ , and  $\vec{E}$  and  $\vec{B}$  are directly observable. Here  $F_{\mu\nu}^A$  is only *covariant* — it rotates into itself under the adjoint action. The non-abelian field strength is therefore not itself an observable; only invariants built from it, like the trace below, are. This is why one must take a trace at all, a step with no counterpart in electromagnetism.

**11 The gauge-invariant trace, and the Dynkin index [p. 15\*\*\*\*\*]**

[p. 15\*\*\*\*\*]

$$\begin{aligned} \text{tr}(F_{\mu\nu}^A T^A F^{B,\mu\nu} T^B) &\longrightarrow \text{tr}(U F_{\mu\nu}^A T^A U^{-1} U F^{B,\mu\nu} T^B U^{-1}) \\ &= \text{tr}(U F_{\mu\nu}^A T^A F^{B,\mu\nu} T^B U^{-1}) = \\ &\stackrel{(*)}{=} \text{tr}(F_{\mu\nu}^A T^A F^{B,\mu\nu} T^B) \end{aligned}$$

(\*) using cyclic property of tr

We thus find that  $\text{tr}(F_{\mu\nu} F^{\mu\nu})$ , with  $F_{\mu\nu} \equiv F_{\mu\nu}^A T^A$ , is a gauge invariant object.

We can also rewrite this object as follows

$$\text{tr}(F_{\mu\nu} F^{\mu\nu}) = \text{tr}(F_{\mu\nu}^A T^A F^{B,\mu\nu} T^B) = F_{\mu\nu}^A F^{B,\mu\nu} \underbrace{\text{tr}(T^A T^B)}_{\text{in principle, we need this trace}} \quad \otimes$$

In the case of a compact Lie group, the exact value of this trace depends on the specific representation of the generators  $T^A$ , and we write

$$\text{tr}(T_R^A T_R^B) = C_R \delta_{AB}$$

$C_R$  is called “Dynkin index” of the representation “ $R$ ”

For instance, in the fundamental representation of  $SU(N)$  the Dynkin index is  $1/2$ .

$$\otimes = F_{\mu\nu}^A F^{B,\mu\nu} C_R \delta_{AB} = C_R F_{\mu\nu}^A F^{A,\mu\nu}$$

[p. 16\*\*\*\*\*]

Irrespectively on the specific value of  $C_R$ , the important point here is that

$$\sum_{A=1}^{N^2-1} F_{\mu\nu}^A F^{A,\mu\nu} \quad \text{is invariant under local } G \text{ transformations}$$

**Why  $\text{tr}(T^A T^B) \propto \delta_{AB}$** 

This is where compactness pays off again. For a compact simple group one can always choose a basis of generators in which  $\text{tr}(T^A T^B) = C_R \delta_{AB}$  — the trace form (Killing form) is positive definite and can be diagonalized and normalized. Two consequences:

- The kinetic term collapses to the simple sum  $\sum_A F_{\mu\nu}^A F^{A,\mu\nu}$ , so each gauge field gets a properly normalized, *same-sign* kinetic term. Positive definiteness is essential: a non-compact group would give some generators negative-norm kinetic terms, i.e. ghosts.
- $\delta_{AB}$  also lets algebra indices be raised and lowered freely, which is why  $f_{ABC}$  can be taken fully antisymmetric (as asserted on [p. 1\*\*\*\*\*]).

For  $SU(N)$  fundamental, the standard normalization is  $\text{tr}(T^A T^B) = \frac{1}{2} \delta_{AB}$ , so  $C_R = 1/2$  — for  $SU(2)$  this is the familiar  $T^A = \sigma^A/2$ . The adjoint representation of  $SU(N)$  has  $C_A = N$  instead. These constants are exactly what appear in the beta function of the theory, and hence in asymptotic freedom.

**12 Computing  $F_{\mu\nu}^A$  explicitly [pp. 16–17\*\*\*\*\*]**

We only need to find  $F_{\mu\nu}^A$ !

We use the definition ■

We compute explicitly the left hand side

$$D_\mu D_\nu \Phi - (\mu \leftrightarrow \nu) = i F_{\mu\nu}^A T^A \Phi$$

$$\begin{aligned} D_\mu D_\nu \Phi &= (\partial_\mu + i G_\mu^A T^A) (\partial_\nu + i G_\nu^A T^A) \Phi = \\ &= (\partial_\mu + i G_\mu^A T^A) (\partial_\nu \Phi + i G_\nu^A T^A \Phi) = \\ &= \partial_\mu \partial_\nu \Phi + i (\partial_\mu G_\nu^A) T^A \Phi + i G_\nu^A T^A (\partial_\mu \Phi) \\ &\quad + i G_\mu^A T^A (\partial_\nu \Phi) - G_\mu^A T^A G_\nu^B T^B \Phi \end{aligned}$$

When we take the difference, we are left with

$$\begin{aligned} D_\mu D_\nu \Phi - D_\nu D_\mu \Phi &= i (\partial_\mu G_\nu^A) T^A \Phi - i (\partial_\nu G_\mu^A) T^A \Phi \\ &\quad - G_\mu^A T^A G_\nu^B T^B \Phi + G_\nu^A T^A G_\mu^B T^B \Phi = \\ &= i (\partial_\mu G_\nu^A - \partial_\nu G_\mu^A) T^A \Phi + G_\nu^A G_\mu^B (T^A T^B - T^B T^A) \Phi = \end{aligned}$$

[p. 17\*\*\*\*\*]

$$\begin{aligned} &= i (\partial_\mu G_\nu^A - \partial_\nu G_\mu^A) T^A \Phi + G_\nu^A G_\mu^B i f_{ABC} T^C \Phi = \\ &= i (\partial_\mu G_\nu^A - \partial_\nu G_\mu^A) T^A \Phi + i G_\nu^C G_\mu^B f_{CBA} T^A \Phi = \\ &= i (\partial_\mu G_\nu^A - \partial_\nu G_\mu^A - f_{ABC} G_\mu^B G_\nu^C) T^A \Phi \end{aligned}$$

From the comparison with the right-hand side

$$F_{\mu\nu}^A = \partial_\mu G_\nu^A - \partial_\nu G_\mu^A - f_{ABC} G_\mu^B G_\nu^C$$

### The extra term, and a check

The first two terms are exactly the abelian field strength, one copy for each value of  $A$ . The third term,  $-f_{ABC} G_\mu^B G_\nu^C$ , is new and is *quadratic in the gauge field*. It exists only because the generators fail to commute.

**The sign chain, since it is easy to lose.** Going from the second to the third line uses full antisymmetry of  $f$ : relabelling  $A \rightarrow C$  in  $G_\nu^A G_\mu^B f_{ABC} T^C$  and then  $C \rightarrow A$  gives  $i f_{CBA} G_\mu^B G_\nu^C$ , and  $f_{CBA} = -f_{ABC}$  because  $ABC \rightarrow CBA$  swaps two indices. Hence the minus sign in the box.

**Why this single term changes everything.**  $F$  is now quadratic in  $G$ , so  $F^2$  contains terms with three and four powers of  $G$ : the Lagrangian  $\sim F^2$  is no longer a free theory. Those are the cubic and quartic self-interactions extracted on [p. 19\*\*\*\*\*–p. 21\*\*\*\*\*]. Set  $f_{ABC} = 0$  and the whole lecture collapses to dim  $G$  non-interacting copies of electromagnetism.

## 13 The Lagrangian, and the rescaling $G_\mu^A \rightarrow g G_\mu^A$ [pp. 17–18\*\*\*\*\*]

We thus arrive at the Lagrangian

$$\mathcal{L} = -\frac{1}{4g^2} F_{\mu\nu}^A F^{A,\mu\nu} + \mathcal{L}(\Phi, D_\mu \Phi)$$

← note that these two terms are independently gauge invariant  
“ $g$ ”, at this stage, is a free constant

Let us look more in detail about the structure of this term.

We write, using the definition

[p. 18\*\*\*\*\*]

$$\begin{aligned} F_{\mu\nu}^A F^{A,\mu\nu} &= \left( \partial_\mu G_\nu^A - \partial_\nu G_\mu^A - f_{ABC} G_\mu^B G_\nu^C \right) \times \\ &\quad \left( \partial^\mu G^{A,\nu} - \partial^\nu G^{A,\mu} - f_{ADE} G^{D,\mu} G^{E,\nu} \right) = \\ &= \underbrace{\left( \partial_\mu G_\nu^A - \partial_\nu G_\mu^A \right) \left( \partial^\mu G^{A,\nu} - \partial^\nu G^{A,\mu} \right)}_{\text{interactions}} + \text{interactions} \end{aligned}$$

This is the same as the term  $F_{\mu\nu} F^{\mu\nu} = (\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu)$  for the photon field but written for  $G_\mu^A$  (“ $A$ ” fixed)

We thus rescale the gauge fields

$$G_\mu^A \rightarrow g G_\mu^A$$

$$\begin{aligned}
-\frac{1}{4g^2} F_{\mu\nu}^A F^{A,\mu\nu} &= -\frac{1}{4g^2} \left( g \partial_\mu G_\nu^A - g \partial_\nu G_\mu^A - g^2 f_{ABC} G_\mu^B G_\nu^C \right) \times \\
&\quad \left( g \partial^\mu G^{A,\nu} - g \partial^\nu G^{A,\mu} - g^2 f_{ADE} G^{D,\mu} G^{E,\nu} \right) = \\
&= -\frac{1}{4g^2} (g^2) \left( \partial_\mu G_\nu^A - \partial_\nu G_\mu^A - g f_{ABC} G_\mu^B G_\nu^C \right) \\
&\quad \left( \partial^\mu G^{A,\nu} - \partial^\nu G^{A,\mu} - g f_{ADE} G^{D,\mu} G^{E,\nu} \right) = \\
&= -\frac{1}{4} \left( \partial_\mu G_\nu^A - \partial_\nu G_\mu^A \right) \left( \partial^\mu G^{A,\nu} - \partial^\nu G^{A,\mu} \right) + \text{interactions}
\end{aligned}$$

With this normalization, each field  $G_\mu^A$  has a photon-like kinetic term.

#### Two conventions, one theory

It is worth being explicit about what just happened, because the placement of  $g$  changes and it is a common source of confusion.

|                | before rescaling                                                        | after $G \rightarrow gG$                                                  |
|----------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------|
| $D_\mu \Phi$   | $\partial_\mu \Phi + i G_\mu^A T^A \Phi$                                | $\partial_\mu \Phi + ig G_\mu^A T^A \Phi$                                 |
| $F_{\mu\nu}^A$ | $\partial_\mu G_\nu^A - \partial_\nu G_\mu^A - f_{ABC} G_\mu^B G_\nu^C$ | $\partial_\mu G_\nu^A - \partial_\nu G_\mu^A - g f_{ABC} G_\mu^B G_\nu^C$ |
| kinetic term   | $-\frac{1}{4g^2} F^A F^A$                                               | $-\frac{1}{4} F^A F^A$                                                    |

Both describe the same physics —  $g$  is just a field normalization away from one place or the other. The second convention is the standard one and the one used from [p. 19\*\*\*\*\*] on, because it makes the free part look exactly like  $\dim G$  copies of Maxwell, so the propagator can be read off immediately.

The first convention has its own virtue: with  $-\frac{1}{4g^2} F^2$ , the coupling appears *only* as an overall factor multiplying the whole gauge action. That makes the classical field equations independent of  $g$ , and it is the form in which the coupling is most naturally seen as  $1/g^2$  counting loops. It is also the form used in lattice gauge theory and in instanton calculations.

## 14 Yang–Mills theory [p. 19\*\*\*\*\*]

We arrive at the so called Yang–Mills theories

[p. 19\*\*\*\*\*]

$$\begin{aligned}
\mathcal{L} &= -\frac{1}{4} F_{\mu\nu}^A F^{A,\mu\nu} + \mathcal{L}_M(\Phi, D_\mu \Phi) \\
F_{\mu\nu}^A &= \partial_\mu G_\nu^A - \partial_\nu G_\mu^A - g f_{ABC} G_\mu^B G_\nu^C \\
D_\mu \Phi &= \partial_\mu \Phi + ig G_\mu^A T^A \Phi
\end{aligned}$$

We distinguish between

$$\mathcal{L}_{YM} \equiv -\frac{1}{4} F_{\mu\nu}^A F^{A,\mu\nu} \quad \text{“pure” Yang–Mills}$$

## ■ Pure Yang–Mills theory $\mathcal{L}_M(\Phi, D_\mu \Phi)$ “matter sector”

We focus on  $\mathcal{L}_{YM} = -\frac{1}{4}F_{\mu\nu}^A F^{A,\mu\nu}$ , and expand it more carefully.

$$\mathcal{L}_{YM} = -\frac{1}{4} \left( \partial_\mu G_\nu^A - \partial_\nu G_\mu^A - g f_{ABC} G_\mu^B G_\nu^C \right) \times \\ \left( \partial^\mu G^{A,\nu} - \partial^\nu G^{A,\mu} - g f_{ADE} G^{D,\mu} G^{E,\nu} \right) =$$

[p. 20\*\*\*\*\*]

$$\begin{aligned} &= -\frac{1}{4} (\partial_\mu G_\nu^A - \partial_\nu G_\mu^A) (\partial^\mu G^{A,\nu} - \partial^\nu G^{A,\mu}) \\ &\quad - \frac{1}{4} (\partial_\mu G_\nu^A - \partial_\nu G_\mu^A) (-g f_{ADE} G^{D,\mu} G^{E,\nu}) \\ &\quad - \frac{1}{4} (-g f_{ABC} G_\mu^B G_\nu^C) (\partial^\mu G^{A,\nu} - \partial^\nu G^{A,\mu}) \\ &\quad - \frac{1}{4} (-g f_{ABC}) (-g f_{ADE}) (G_\mu^B G_\nu^C) (G^{D,\mu} G^{E,\nu}) = \\ &= -\frac{1}{4} (\partial_\mu G_\nu^A - \partial_\nu G_\mu^A) (\partial^\mu G^{A,\nu} - \partial^\nu G^{A,\mu}) \\ &\quad + \frac{g}{2} f_{ABC} (\partial_\mu G_\nu^A - \partial_\nu G_\mu^A) G^{B,\mu} G^{C,\nu} \\ &\quad - \frac{g^2}{4} f_{ABC} f_{ADE} G_\mu^B G_\nu^C G^{D,\mu} G^{E,\nu} = \\ &= -\frac{1}{4} (\partial_\mu G_\nu^A - \partial_\nu G_\mu^A) (\partial^\mu G^{A,\nu} - \partial^\nu G^{A,\mu}) \\ &\quad + g f_{ABC} (\partial_\mu G_\nu^A) G^{B,\mu} G^{C,\nu} - \frac{g^2}{4} f_{ABC} f_{ADE} G_\mu^B G_\nu^C G^{D,\mu} G^{E,\nu} \end{aligned}$$

$$\mathcal{L}_{YM} = \underbrace{-\frac{1}{4} (\partial_\mu G_\nu^A - \partial_\nu G_\mu^A) (\partial^\mu G^{A,\nu} - \partial^\nu G^{A,\mu})}_{\text{free theory}} \\ + \underbrace{g f_{ABC} (\partial_\mu G_\nu^A) G^{B,\mu} G^{C,\nu} - \frac{g^2}{4} f_{ABC} f_{ADE} G_\mu^B G_\nu^C G^{D,\mu} G^{E,\nu}}_{\text{interactions}}$$

## 15 The three terms individually [pp. 21–22\*\*\*\*\*]

[p. 21\*\*\*\*\*]

Individually

■  $-\frac{1}{4} (\partial_\mu G_\nu^A - \partial_\nu G_\mu^A) (\partial^\mu G^{A,\nu} - \partial^\nu G^{A,\mu})$

kinetic term

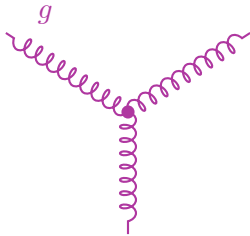
$$[G_\mu^A] = [M]$$

Each gauge boson  $G_\mu^A$  propagates like a photon


$$\frac{-ig_{\mu\nu}}{p^2 + i\epsilon}$$

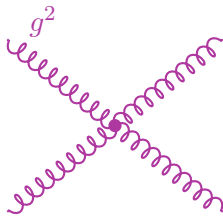
$$gf_{ABC}(\partial_\mu G_\nu^A)G^{B,\mu}G^{C,\nu}$$

trilinear self interaction



$$-\frac{g^2}{4}f_{ABC}f_{ADE}G_\mu^BG_\nu^CG^{D,\mu}G^{E,\nu}$$

quartic self interaction



In summary:

$\mathcal{L}_{YM}$  describes an interacting field theory of dim  $G$  massless gauge bosons

There is (in the case of simple group  $G$ ) one parameter, “ $g$ ”, called the “gauge coupling” that controls the strenght of the interactions.

[p. 22\*\*\*\*\*]

“ $g$ ” is dimensionless  $\rightsquigarrow$  interactions are renormalizable

The theory is non-linear, but non-linearities stop at the quartic order

Comparing with gravity — why this theory is well behaved

This is worth pausing on, because the contrast with Lectures 19–21 is exact and instructive. Both theories are non-linear for the same reason (the gauge field carries the charge it couples to), yet they behave completely differently as quantum theories:

|                       | Yang–Mills               | gravity                           |
|-----------------------|--------------------------|-----------------------------------|
| coupling              | $g$ , dimensionless      | $\kappa$ , $[\kappa] = [M]^{-1}$  |
| self-interactions     | stop at quartic          | infinite tower $\mathcal{O}(h^n)$ |
| amplitude at high $E$ | $\sim g^2$ , no growth   | $\sim E^2/M_P^2$ , grows          |
| status                | renormalizable, complete | effective theory, cutoff $M_P$    |

**Why the series terminates here but not for gravity.**  $F$  is quadratic in  $G$ , so  $F^2$  is quartic and stops. In gravity, consistency required the energy–momentum tensor of the gravitational field itself to be included, and that fed back into an infinite series. The technical marker of the difference is the dimension of the coupling: dimensionless couplings give operators of dimension exactly 4, and there are finitely many of those; a coupling of negative dimension permits an unbounded tower of higher-dimension operators.

**Why**  $[G_\mu^A] = [M]$ . From the kinetic term  $(\partial G)^2$  with  $[\mathcal{L}] = [M]^4$  and  $[\partial] = [M]$ . Then the cubic vertex  $gG(\partial G)G$  has dimension 4 only if  $[g] = 0$  — the same counting that makes the theory renormalizable.

The one loose end this lecture does not reach: quantizing a non-abelian gauge theory requires more care than QED (Faddeev–Popov ghosts), and the resulting coupling *runs*, decreasing at high energy. That asymptotic freedom is what makes QCD calculable at short distances, and it is a direct consequence of the self-interactions drawn above.

## 16 The matter sector (fermionic case) [pp. 22–23\*\*\*\*\*]

### ■ Matter sector (fermionic case)

As a benchmark example, we consider a fermionic matter Lagrangian with Dirac fields in the fundamental, considering  $G = SU(N)$  as gauge group.

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^A F^{A,\mu\nu} + \bar{\Psi}i\gamma^\mu(D_\mu\Psi) - m\bar{\Psi}\Psi$$

We open up the covariant derivative

$$\bar{\Psi}i\gamma^\mu\partial_\mu\Psi - g\bar{\Psi}\gamma^\mu G_\mu^A T^A\Psi$$

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^A F^{A,\mu\nu} + \bar{\Psi}i\gamma^\mu\partial_\mu\Psi - m\bar{\Psi}\Psi - g\bar{\Psi}\gamma^\mu G_\mu^A T^A\Psi$$

↖ interaction between the gauge bosons and the fermions, with strenght controlled by the same coupling  $g$

[p. 23\*\*\*\*\*]

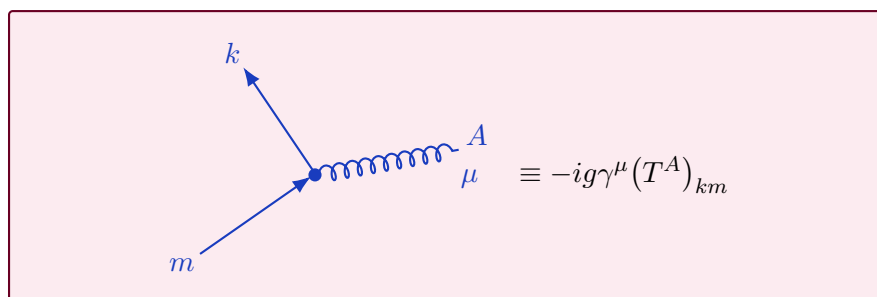
In components, the interaction is

$$-g\bar{\Psi}^k\gamma^\mu G_\mu^A (T^A)_{km}\Psi^m$$

↖ destroy a fermion of type “ $m$ ”,  
create an anti-fermion of type “ $m$ ”

↙ destroy an anti-fermion of type “ $k$ ”,  
create a fermion of type “ $k$ ”

The Feynman rule is



↔ Important observation

By exchanging a non-abelian gauge boson, it is possible to “change” the type of fermions

### The last observation is the physically decisive one

In QED the vertex is  $-iq\gamma^\mu$  — a number times a Dirac matrix. Here it is  $-ig\gamma^\mu(T^A)_{km}$ , and  $T^A$  is an *off-diagonal* matrix in general. So the incoming fermion of type  $m$  and the outgoing fermion of type  $k$  need not be the same type: emitting a gauge boson rotates the particle within its multiplet.

Concretely:

- In QCD, a red quark emitting a gluon becomes a blue quark; the gluon carries away the colour difference. This is why gluons are coloured, and it is the same statement as “the gauge field transforms in the adjoint”.
- In the electroweak theory, a  $W$  boson converts an electron into a neutrino, or an up quark into a down quark. That is precisely why the weak interaction changes flavour, and hence why the neutron decays.

Everything traces back to one algebraic fact: for a non-abelian group the generators are matrices that do not commute, and non-commuting matrices cannot all be diagonal. Note also that a *single* coupling  $g$  controls the gauge-boson self-interactions and the gauge-boson-matter interaction — the gauge principle leaves no freedom to make them differ, which is a sharp, testable prediction.

## 17 The case of semi-simple Lie groups [p. 24\*\*\*\*\*]

[p. 24\*\*\*\*\*]

### ■ The case of semi-simple Lie groups

We now consider the case in which the group  $G$  is semi-simple

$$G = G_1 \times G_2 \times \cdots \times G_m$$

direct product of simple Lie group

From the Lie algebra viewpoint

$$\dim G = \dim G_1 + \cdots + \dim G_m$$

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \cdots \oplus \mathfrak{g}_m$$

direct sum of Lie subalgebras

We work out the same construction outlined in the case of simple groups. In particular:

We introduce dim  $G$  gauge bosons  $G_\mu^A$  in the covariant derivative

$$D_\mu \Phi = \partial_\mu \Phi + i G_\mu^A T^A \Phi$$

with field strength

$$F_{\mu\nu}^A = \partial_\mu G_\nu^A - \partial_\nu G_\mu^A - f_{ABC} G_\mu^B G_\nu^C$$

As in the previous case,  $G_\mu^A$  and  $F_{\mu\nu}^A$  transform under the adjoint representation of the group  $G$ .

### Where the scan stops — and the one point it was heading for

The sheet ends here. The construction goes through essentially unchanged, but there is one genuinely new feature that the semi-simple case introduces, and it is worth stating because it is what the section exists for.

**One coupling per simple factor.** For a *simple* group, gauge invariance forces a single coupling  $g$ . For a product  $G = G_1 \times \cdots \times G_m$ , the structure constants vanish between different factors ( $f_{ABC} = 0$  unless  $A, B, C$  all belong to the same  $\mathfrak{g}_i$ ), so each factor's field strength is separately gauge invariant and each can carry its own normalization:

$$\mathcal{L}_{YM} = - \sum_{i=1}^m \frac{1}{4g_i^2} F_{\mu\nu}^{A_i} F^{A_i, \mu\nu},$$

i.e.  $m$  independent couplings  $g_1, \dots, g_m$ . Nothing relates them — which is exactly why grand unification, which *would* relate them by embedding  $G$  in a single simple group, is an interesting hypothesis rather than a theorem.

**The example that matters.** The Standard Model gauge group is

$$G = SU(3)_C \times SU(2)_L \times U(1)_Y, \quad \dim G = 8 + 3 + 1 = 12,$$

with three couplings  $g_3, g_2, g_1$  — twelve gauge bosons: eight gluons, and the four that become  $W^\pm, Z$  and the photon after electroweak symmetry breaking. (Strictly,  $U(1)$  is abelian rather than simple, so this group is “reductive” rather than semi-simple in the mathematician’s sense; the physics construction is identical, with  $f_{ABC} = 0$  for the  $U(1)$  factor and hence no self-interaction for its gauge boson.)

**What has been achieved.** Starting from nothing but “a free Lagrangian with a global compact symmetry”, the three steps of this lecture produced: a specific number of massless vector bosons ( $\dim G$ ), their self-interactions (fixed by  $f_{ABC}$ ), their coupling to matter (fixed by the representation), and a single coupling constant controlling all of it. This is the same gauge-principle logic that gave QED in earlier lectures and general coordinate invariance in Lecture 19 — applied here it gives the strong and electroweak interactions.

### Summary of Lecture 22 — the logic in nine lines

1. **Step 1.** Start from a free Lagrangian invariant under a *global* compact Lie group  $G$ , with  $\Phi \rightarrow U\Phi$ ,  $U = \exp(-i\theta_A T^A)$  and  $[T^A, T^B] = if_{ABC}T^C$ . Compactness makes the generators Hermitian and  $f_{ABC}$  fully antisymmetric.
2. **Step 2.** Let  $\theta_A \rightarrow \theta_A(x)$ . The derivative term breaks, because  $\partial_\mu(U\Phi)$  produces the stray piece  $(\partial_\mu U)\Phi$ .
3. Introduce  $\dim G$  gauge fields via  $D_\mu\Phi = \partial_\mu\Phi + iG_\mu^A T^A\Phi$  and demand  $D_\mu\Phi \rightarrow U(D_\mu\Phi)$ . This forces  $G_\mu \rightarrow i(\partial_\mu U)U^{-1} + UG_\mu U^{-1}$ .
4. The second term survives for constant  $U$ : unlike the photon, the gauge bosons are charged. Infinitesimally they transform with generators  $(t^B)_{AC} = -if_{BAC}$  — the *adjoint representation*.
5. **Step 3.** Define the field strength through the commutator of covariant derivatives,  $[D_\mu, D_\nu]\Phi = iF_{\mu\nu}^A T^A\Phi$ . It then transforms homogeneously,  $F \rightarrow UFU^{-1}$ , so  $\text{tr}(F_{\mu\nu}F^{\mu\nu})$  is gauge invariant.
6. Computing it gives  $F_{\mu\nu}^A = \partial_\mu G_\nu^A - \partial_\nu G_\mu^A - f_{ABC}G_\mu^B G_\nu^C$  — the abelian expression plus a term quadratic in  $G$ .

7. Rescaling  $G \rightarrow gG$  puts the Lagrangian in standard form,  $\mathcal{L} = -\frac{1}{4}F^A F^A + \mathcal{L}_M$ , with  $g$  appearing in both  $F$  and  $D_\mu$ .
8. Expanding  $F^2$  gives  $\dim G$  photon-like kinetic terms plus cubic and quartic *self*-interactions. Since  $[g] = 0$  the theory is renormalizable and the non-linearities terminate at quartic order — unlike gravity.
9. Matter couples as  $-g\bar{\Psi}^k \gamma^\mu G_\mu^A (T^A)_{km} \Psi^m$ . Because  $T^A$  is off-diagonal, emitting a gauge boson can change the type of fermion: colour change in QCD, flavour change in the weak interaction.